Introduction to Machine Learning

Machine learning is a field of artificial intelligence in which computer systems learn patterns from data.

Supervised learning uses labeled examples. A model receives inputs and known outputs during training. Classification predicts categories, such as spam or not spam. Regression predicts a number, such as house price.

Key terms:
- Dataset: a collection of examples.
- Feature: an input variable used by a model.
- Label: the correct output in supervised learning.

Training, Testing, and Evaluation

A dataset is commonly divided into training data and test data. The model learns patterns from the training data. The test data checks whether the model can work on unseen examples.

Accuracy is the proportion of correct predictions. Precision measures how many predicted positive cases were actually positive. Recall measures how many actual positive cases were found by the model.

Overfitting happens when a model memorizes training data but performs poorly on new data.

Naive Bayes

Naive Bayes is a probabilistic classification method. It uses Bayes' rule to estimate which class is more likely.

The word naive means the model assumes features are conditionally independent given the class. This assumption is often not perfectly true, but the method can still work well for text classification.

Example: an email classifier can use words such as free, offer, and meeting to estimate whether an email is spam.